26900403

Discipline Specific Elective (DSE) MG5DSEPES305 – Sports Nutrition Essentials

MGU-UGP (Honours) | BPES | Fifth Semester | October 2026

ANSWER KEY

Part A – Multiple-Choice Questions

Answer all questions. Each question carries 1 mark.

1. How can adequate protein intake contribute to athletic performance?

Answer: A. By supporting muscle protein synthesis

2. Which nutrient provides a significant amount of energy during lower-intensity, long-duration exercise?

Answer: A. Fat

3. When evaluating different psychological strategies for athlete weight management, which should be prioritized as most sustainable long-term?

Answer: C. Strategies that build resilience and healthy coping skills rather than quick-fix or appearance-focused approaches

4. Reduced pre-exercise anxiety and improved confidence relate mainly to:

Answer: D. Psychological readiness

5. The main purpose of recovery nutrition is to:

Answer: A. Restore glycogen, repair muscle and replace fluids

6. Which category of sports is most closely associated with pressure to make a specific weight class before competition?

Answer: C. Weight-category sports such as wrestling, boxing, and lightweight rowing

7. Which of these is a practical rehydration option?

Answer: D. Buttermilk (sambaram)

8. A small carbohydrate snack 30–60 minutes before exercise may provide approximately:

Answer: B. 30–50 g carbohydrate

9. Which of the following foods is a good source of calcium?

Answer: B. Milk

10. Adequate carbohydrate also has a protein-sparing effect because it:

Answer: A. Reduces the use of amino acids for energy

Part B – Short-Answer Questions

Answer any five questions. Each question carries 2 marks.

11. Why are pre- and post-training meals important?

Pre-training meals provide energy for exercise, improve performance, and help prevent fatigue. Post-training meals replenish glycogen stores, repair damaged muscles, promote muscle growth, and support recovery through adequate carbohydrate, protein, and fluid intake.

12. Why is resistance training essential alongside a caloric surplus?

Resistance training is essential alongside a caloric surplus because it stimulates muscle growth and muscle protein synthesis. It increases muscle strength and power, supports bone health, improves body composition, and helps direct excess calories towards muscle development while minimizing excessive fat gain.

13. Why should fluid intake be combined with carbohydrate and protein after exercise?

Fluid intake should be combined with carbohydrates and protein after exercise to restore lost fluids and electrolytes, replenish depleted glycogen stores, and repair damaged muscle tissues. This combination promotes muscle recovery, reduces fatigue, and prepares the body for the next training session.

14. What is the leucine threshold?

The leucine threshold is the amount of leucine, an essential branched-chain amino acid (BCAA), needed in a meal to sufficiently stimulate muscle protein synthesis. It is generally around 2–3 grams per meal in healthy adults, supporting muscle growth and repair.

15. Why can dehydration before competition impair performance?

Dehydration before competition can reduce blood volume, increase body temperature, and cause early fatigue. It can also decrease endurance, muscle strength, concentration, and coordination, leading to reduced sports performance.

16. Explain the hydration plan recommended alongside pre-exercise meals.

The hydration plan alongside pre-exercise meals involve drinking approximately 5–7 mL of water per kilogram of body weight about 4 hours before exercise to ensure adequate hydration. If urine remains dark or urination is insufficient, an additional 3–5 mL/kg about 2 hours before exercise may be consumed. This helps maintain fluid balance, regulate body temperature, prevent dehydration, and improve sports performance.

17. Why should a healthy weight range be individualized for each athlete?

A healthy weight range should be individualized for each athlete because body composition, height, age, sex, sport-specific demands, muscle mass, and performance goals vary among individuals. Individualized weight management helps maintain optimal performance, prevent nutritional deficiencies and injuries, and support overall health and recovery.

Part C – Short-Essay Questions

Answer any four questions. Each question carries 5 marks.

18. Explain the concept and purpose of carbohydrate loading.

- Carbohydrate loading is a nutritional strategy used to maximize muscle and liver glycogen stores before prolonged endurance events.
- It increases glycogen availability and delays fatigue during prolonged exercise.
- It helps maintain exercise intensity and supports endurance performance.
- It usually involves increased carbohydrate intake before competition, along with an appropriate reduction in training load.

19. Discuss the role of sports nutrition in providing energy for physical activity and supporting muscle growth and repair.

- Carbohydrates provide energy and replenish glycogen stores.
- Fats supply energy, particularly during low-intensity, prolonged exercise.
- Proteins support muscle growth, repair and recovery.
- Vitamins and minerals help regulate metabolism and support normal body functions.
- Water and electrolytes maintain hydration, temperature regulation and muscle function.

20. Explain why calorie-dense whole foods are preferred over processed foods for weight gain.

- Calorie-dense whole foods provide energy along with essential nutrients.
- They support gradual and healthy weight gain and supply protein, healthy fats, vitamins and minerals.
- When combined with resistance training, they can support muscle growth.
- They provide fibre and support digestive health.
- Examples include nuts, seeds, milk, eggs, whole grains, avocados and nut butters.

21. Explain the "front-loading" and "shifting to liquid forms" strategies and how they work together for an event.

- Front-loading means consuming a substantial portion of required energy and nutrients earlier in the day or before the event.
- Shifting to liquid forms means using drinks, smoothies or other suitable liquid foods when solid foods become difficult to tolerate or consume.
- Front-loading helps meet energy needs early.

- Liquid nutrition can make it easier to obtain additional energy and fluids closer to competition.
- Together, these strategies can help meet nutritional requirements while minimizing gastrointestinal discomfort.

22. Explain the general principles of hydration for athletes.

- Begin exercise adequately hydrated.
- Drink fluids regularly according to individual requirements.
- Consider sweat rate, exercise duration, temperature and humidity.
- Replace fluid and electrolyte losses when necessary.
- Monitor changes in body weight and urine colour as practical indicators, and avoid excessive fluid intake.

23. Explain the general principles of hydration for athletes, including the balance between preventing dehydration and overhydration.

- Athletes should replace appropriate amounts of water and electrolytes lost through sweat.
- Drink according to individual needs and environmental conditions.
- Avoid excessive fluid loss, which may impair performance.
- Avoid drinking beyond fluid losses, which can cause exercise-associated hyponatremia.
- Use body-weight changes before and after exercise to help estimate fluid loss and adjust hydration plans according to sweat rate and exercise duration.

Part D – Essay-Type Questions

Answer any two questions. Each question carries 15 marks.

24. Explain pre-exercise, during-exercise and post-exercise hydration strategies for endurance athletes.

Pre-exercise hydration

- Athletes should begin exercise in a well-hydrated state to prevent early dehydration and maintain performance.
- They should drink approximately 5–7 mL of fluid per kilogram of body weight about 4 hours before exercise.
- If urine output is low or urine remains dark, an additional 3–5 mL/kg about 2 hours before exercise may be consumed.
- Water is generally sufficient for shorter activities, while electrolyte-containing drinks may be useful in hot conditions or during prolonged endurance events.
- Athletes should avoid excessive fluid intake immediately before exercise to prevent stomach discomfort and frequent urination.

2. During-exercise hydration

- Athletes should drink fluids at regular intervals according to thirst, sweat rate, exercise intensity and environmental conditions.
- The aim is to replace fluid losses and minimize excessive dehydration during prolonged exercise.
- Sports drinks containing carbohydrates and electrolytes are useful during endurance activities lasting more than one hour because they provide energy and replace sodium losses.
- Athletes should develop an individualized hydration plan based on sweat rate, body weight changes and previous training experience.
- Excessive drinking should be avoided because it can cause exercise-associated hyponatremia, a dangerous decrease in blood sodium concentration.
- Hydration strategies should be practised during training before being used in competition.

3. Post-exercise hydration

- Athletes should replace the fluids lost through sweating to restore normal hydration levels.
- Body weight before and after exercise can help estimate fluid loss; each kilogram of body weight lost represents approximately one litre of net fluid loss.
- Athletes may consume approximately 1.25–1.5 litres of fluid for every kilogram of body weight lost when rapid and complete rehydration is required.
- Sodium-containing fluids or meals help retain consumed fluid and replace sodium lost through sweat.
- Carbohydrate-containing drinks and foods help replenish glycogen stores, while protein supports muscle repair and recovery.
- Athletes should continue drinking fluids with meals until hydration is restored, while avoiding forced or excessive fluid consumption.

Proper pre-exercise, during-exercise and post-exercise hydration helps endurance athletes maintain performance, regulate body temperature, prevent dehydration and support effective recovery. Hydration plans should be individualized according to sweat rate, exercise duration, environmental conditions and the athlete's needs.

25. Discuss the nutritional principles underlying weight management in athletes, covering energy balance, energy density, macronutrient distribution, and the role of resistance training in preserving lean mass during a caloric deficit.

1. Energy Balance

- Energy balance refers to the relationship between calories consumed through food and calories expended through physical activity and body functions.
- A caloric deficit occurs when energy expenditure exceeds energy intake, resulting in weight loss.
- A caloric surplus occurs when energy intake exceeds expenditure, leading to weight gain.
- Athletes should maintain a moderate caloric deficit to reduce body fat while minimizing muscle loss and performance decline.
- Adequate energy intake is essential to support training, recovery and overall health.

2. Energy Density

- Energy density refers to the number of calories present per unit weight of food.
- Low-energy-density foods, such as vegetables, fruits and soups, provide fewer calories relative to their weight and help promote fullness.
- High-energy-density foods, such as fried foods, sweets and foods rich in added fats, provide more calories in smaller portions.
- Choosing nutrient-dense foods helps athletes meet their vitamin, mineral and fibre requirements while controlling calorie intake.
- Appropriate portion sizes and healthy food choices support effective weight management.

3. Macronutrient Distribution

- Carbohydrates: Provide energy for training and replenish muscle glycogen stores. Intake should be adjusted according to training intensity and duration.
- Proteins: Support muscle repair, growth and maintenance. Adequate protein intake is particularly important during a caloric deficit to reduce muscle loss.
- Fats: Provide essential fatty acids, support hormone production and facilitate the absorption of fat-soluble vitamins.
- Athletes should distribute macronutrients according to their sport, training demands, body composition goals and total energy requirements.
- A balanced diet containing adequate carbohydrates, proteins and healthy fats helps maintain performance and overall health.

4. Role of Resistance Training in Preserving Lean Mass

- Resistance training stimulates muscle protein synthesis and helps preserve muscle mass during weight loss.

- It provides a mechanical stimulus that reduces the loss of lean tissue during a caloric deficit.
- Combined with adequate protein intake, resistance training improves the likelihood of losing fat while retaining muscle.
- Athletes should maintain appropriate resistance-training intensity and adjust training volume according to recovery and energy availability.
- Excessively aggressive calorie restriction should be avoided because it may increase muscle loss, fatigue and injury risk.
- Monitoring body composition, strength and performance helps assess the effectiveness of the weight-management programme.

Effective weight management in athletes requires a controlled energy balance, appropriate food choices, balanced macronutrient distribution and regular resistance training. A moderate caloric deficit combined with adequate protein and sufficient recovery helps reduce body fat while preserving lean muscle mass and athletic performance.

26. Discuss the importance of proper nutrition in promoting health, development, tissue maintenance and immune function.

1. Promotion of Health

- Proper nutrition provides essential nutrients required for normal body functioning and overall well-being.
- A balanced diet supplies carbohydrates, proteins, fats, vitamins, minerals and water in appropriate amounts.
- It reduces the risk of nutritional deficiencies and chronic diseases such as obesity, diabetes and cardiovascular diseases.
- Adequate nutrition supports physical fitness, mental health and daily activities.

2. Growth and Development

- Proteins are essential for the growth and development of muscles and other body tissues.
- Calcium, phosphorus and vitamin D support the development of strong bones and teeth.
- Carbohydrates and fats provide energy required for growth, physical activity and normal body functions.
- Vitamins and minerals support brain development, hormonal functions and proper maturation.
- Adequate nutrition is especially important during childhood, adolescence and pregnancy.

3. Tissue Maintenance and Repair

- Proteins provide amino acids required for the repair and replacement of damaged tissues.
- Carbohydrates and fats supply energy needed for tissue maintenance and recovery.
- Vitamin C supports collagen formation and wound healing.
- Zinc promotes cell growth, tissue repair and wound healing.
- Adequate fluid intake supports nutrient transport, waste removal and normal cellular functions.

4. Immune Function

- Proteins are required for the production of antibodies and other immune system components.
- Vitamin C supports immune cell function and protects cells against oxidative damage.
- Vitamin A maintains the integrity of the skin and mucous membranes, which act as protective barriers against infections.
- Vitamin D supports the regulation of immune responses.
- Zinc and iron are essential for normal immune cell development and function.
- Adequate energy and nutrient intake help the body resist infections and recover from illness.

Proper nutrition is essential for maintaining health, supporting growth and development, repairing body tissues and strengthening immune function. A balanced diet containing all essential nutrients in adequate amounts promotes overall well-being and reduces the risk of nutritional deficiencies and diseases.

27. As a coach, design a checklist of “dos and don'ts” for athletes to avoid common nutritional mistakes during pre-exercise, during-exercise and post-exercise phases.

. Pre-exercise Phase

Dos

- Eat a balanced meal rich in carbohydrates 2–4 hours before exercise to provide sufficient energy.
- Include moderate amounts of protein to support muscle function.
- Drink adequate fluids before exercise to begin the session well hydrated.
- Choose familiar, easily digestible foods to prevent stomach discomfort.
- Adjust meal timing and portion sizes according to exercise intensity and duration.

Don'ts

- Do not exercise on an empty stomach if it causes weakness or poor performance.

- Avoid heavy, high-fat and high-fibre meals immediately before exercise.
- Do not try unfamiliar foods or supplements before competition.
- Avoid excessive caffeine intake and energy drinks.
- Do not neglect pre-exercise hydration.

2. During-exercise Phase

Dos

- Drink fluids regularly according to thirst, sweat rate and environmental conditions.
- Use carbohydrate-containing sports drinks or foods during prolonged endurance exercise when appropriate.
- Replace electrolytes, particularly sodium, during prolonged exercise with substantial sweat losses.
- Follow an individualized hydration and fueling plan practised during training.
- Monitor fatigue, thirst and other signs of dehydration.

Don'ts

- Do not ignore thirst or significant fluid losses.
- Avoid excessive fluid intake, which may cause exercise-associated hyponatremia.
- Do not consume large quantities of food that may cause gastrointestinal discomfort.
- Avoid experimenting with new sports drinks or supplements during competition.
- Do not rely on energy drinks as a substitute for proper hydration and nutrition.

3. Post-exercise Phase

Dos

- Consume carbohydrates to replenish muscle glycogen stores, especially after prolonged or intense exercise.
- Include protein to support muscle repair and recovery.
- Replace fluid losses using water and, when necessary, sodium-containing drinks or foods.
- Eat a balanced recovery meal containing carbohydrates, protein, vitamins and minerals.
- Begin recovery nutrition promptly when another training session or competition is scheduled soon.

Don'ts

- Do not skip meals after strenuous exercise.
- Avoid relying exclusively on protein supplements while neglecting carbohydrates and other nutrients.
- Do not ignore fluid and electrolyte losses after heavy sweating.
- Avoid excessive alcohol consumption, which can impair recovery and rehydration.
- Do not overconsume supplements or follow unverified recovery diets.

Coaches should educate athletes to follow appropriate nutrition and hydration practices before, during and after exercise. Avoiding common nutritional mistakes helps maintain energy levels, prevent dehydration, support muscle recovery and improve athletic performance.